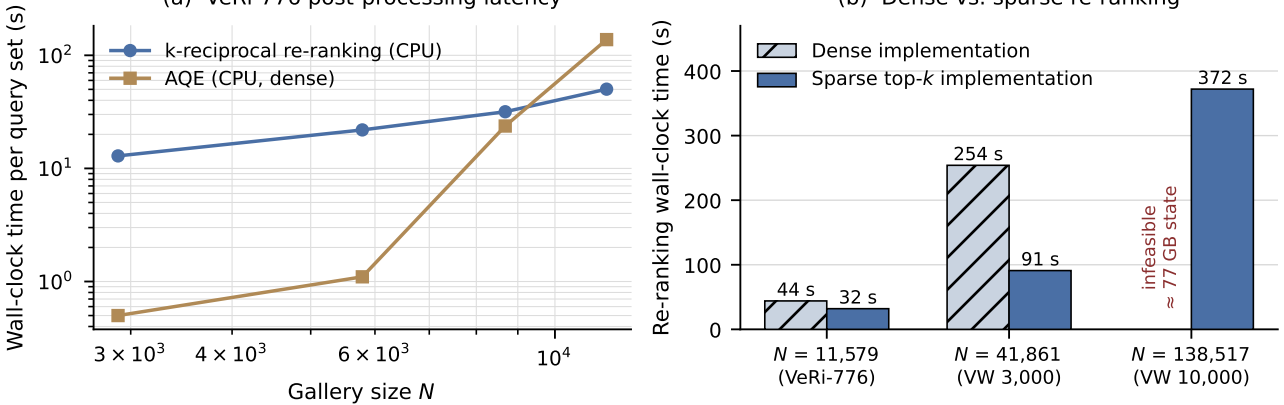
(a) VeRi-776 post-processing latency
k-reciprocal re-ranking (CPU)
AQE (CPU, dense)
Wall-clock time per query set (s)
10^2
10^1
10^0
3×10^3
4×10^3
6×10^3
10^4
Gallery size N
(b) Dense vs. sparse re-ranking
Dense implementation
Sparse top-k implementation
Re-ranking wall-clock time (s)
400
300
200
100
0
44 s
32 s
254 s
91 s
372 s
infeasible
≈ 77 GB state
N = 11,579
(VeRi-776)
N = 41,861
(VW 3,000)
N = 138,517
(VW 10,000)